

Enterprise DFIR Field Guide

**Windows Endpoints, Memory, Active Directory, and Hybrid/
Cloud Identity**

Contents

1. Enterprise DFIR Field Guide	3
1.1 How this guide is organized	3
1.2 Scope	3
1.3 The modules	3
1.4 A note on sources	4
2. Glossary	5
2.1 Referenced From	6

1. Enterprise DFIR Field Guide

An interactive, data-driven reference for enterprise digital forensics and incident response — built for defenders working across Windows endpoints, memory, Active Directory, and hybrid/cloud identity.

This is not written for criminal-prosecution workflows. It's built for the SOC analyst, incident responder, and threat hunter who needs to answer one question fast: **is this normal, or is this an incident?**

1.1 How this guide is organized

Three layers, built to work together:

Knowledge Base (Modules 0–7) — one canonical page per artifact, log source, or mechanism. Each page follows the same shape: what it is, what normal looks like, what a red flag looks like, how to collect it, and how it maps to [MITRE ATT&CK](#).

Case Studies — full, narrated investigations showing the reasoning *between* the reference pages: why one finding sends you to check a specific next thing, how you catch a timestamp lying to you, how much corroboration is actually enough. The Knowledge Base teaches recognition; these teach synthesis.

Investigation Playbooks (Module 8) — scenario-driven workflows (Business Email Compromise, ransomware, insider threat, domain compromise, data exfiltration...) that link into the Knowledge Base rather than repeating it.

New to DFIR? Start at [Module 0: Foundations](#) — everything downstream assumes the vocabulary and frameworks introduced there, including [Timeline Construction & Correlation](#), which the Case Studies lean on directly.

Already investigating something specific? Jump straight to the relevant page in [Investigation Playbooks](#).

1.2 Scope

In scope	Not covered here
Windows workstations, laptops, servers, VMs	macOS / Linux endpoint forensics
Active Directory & Domain Controllers	Non-Microsoft cloud (AWS/GCP) IAM forensics
Windows memory forensics	Mobile device forensics
Entra ID, hybrid identity, Microsoft 365	Network packet-level forensics (see SANS FOR572)
Microsoft Defender suite (AV through XDR)	Legal / chain-of-custody procedure for litigation

1.3 The modules

1. [Foundations](#) — methodology and shared frameworks
2. [Windows Endpoint Forensics](#) — filesystem, registry, event logs, process trees
3. [Active Directory & Domain Controllers](#)
4. [Windows Memory Forensics](#)
5. [PowerShell Forensics](#) — logging, obfuscation, decoding, fileless execution
6. [Persistence Catalog](#) — every mechanism, endpoint and cloud, ATT&CK-tagged
7. [Cloud Identity \(Entra ID / Hybrid\)](#)
8. [Microsoft Defender Suite for IR](#)
9. [Network & Perimeter Log Analysis](#) — DNS and proxy/firewall log triage (not full packet forensics — that's out of scope by design)
10. [Anti-Forensics & Data Recovery](#) — what attackers do to destroy or hide evidence, and what's still recoverable anyway

- 11. [Case Studies](#) — full narrated investigations connecting the dots across modules
- 12. [Investigation Playbooks](#) — the scenario layer
- 13. [Practice Drills](#) — hands-on exercises with simulated data, answers included

Two standing references, useful throughout rather than tied to one module: the [Windows IR Quick Reference](#) (a single dense page for an active incident) and the [Glossary](#).

1.4 A note on sources

Every page here is original writing, built from the facts, structures, and taxonomies published by SANS (course material and posters), Microsoft Learn/MSTIC, and the wider DFIR community (including practitioners like 13cubed). Nothing is reproduced verbatim — each page links to primary sources for readers who want the original.

🕒 August 3, 2026

2. Glossary

A single lookup point for the acronyms and terms used throughout this guide. Each entry links to the page that covers it in depth — this page defines, it doesn't re-teach.

ADS (Alternate Data Stream) — an NTFS feature allowing a file to carry additional, named, normally-invisible data streams. See [Alternate Data Streams](#).

AIR (Automated Investigation and Remediation) — Defender for Endpoint's capability to automatically investigate and, for high-confidence verdicts, remediate alerts. See [Automated Investigation & Remediation](#).

AMSI (Antimalware Scan Interface) — the interface letting AV/EDR engines inspect script content immediately before execution, after any obfuscation has already been unwound. See [Evasion Detection](#).

AD CS (Active Directory Certificate Services) — AD's built-in PKI; misconfigurations here (the "ESC" family) allow certificate-based authentication as another user. See [AD CS Abuse](#).

ATT&CK — MITRE's knowledge base of adversary tactics and techniques, used throughout this guide as a tagging system. See [ATT&CK Primer](#).

Body file — a standardized intermediate format (MACtime) for filesystem timeline data, consumable by `plaso`. See [Timeline Construction & Correlation](#).

CLM (Constrained Language Mode) — a PowerShell execution restriction blocking direct .NET/COM access, commonly paired with AppLocker/WDAC. See [Evasion Detection](#).

DCSync — abuse of the AD replication protocol to request password hashes for any account without executing on a Domain Controller. See [DCSync Detection](#).

DGA (Domain Generation Algorithm) — malware technique generating many candidate C2 domains algorithmically to defeat static blocklisting. See [DNS Analysis](#).

DKOM (Direct Kernel Object Manipulation) — a rootkit technique unlinking a process from the OS's own tracking structures to hide it from list-based enumeration. See [Process Analysis](#).

EPROCESS — the Windows kernel structure representing a running process. See [EPROCESS Internals](#).

ESC1 / ESC4 / ESC8 — specific AD CS misconfiguration classes enabling certificate-based privilege escalation. See [AD CS Abuse](#).

ETHREAD — the kernel structure representing a thread; its start address is the key forensic field. See [Thread Analysis](#).

EVTX — the Windows Event Log file format. See [Event Log Key IDs Reference](#) and, for recovery after tampering, [Log & Artifact Recovery](#).

FILETIME — the underlying Windows timestamp format: 100-nanosecond intervals since January 1, 1601 UTC. See [Timeline Construction & Correlation](#).

gMSA (Group Managed Service Account) — a service account type with automatically-rotated, unknowable passwords, recommended for high-privilege service accounts. Referenced in [Kerberoasting](#) mitigations and [Entra Connect as Target](#).

IOC (Indicator of Compromise) — a specific, low-durability artifact of an attack (a hash, domain, IP). See [Pyramid of Pain](#) for why these rank lowest in detection value.

JA3 / JA4 — TLS client fingerprinting techniques; JA3 is now considered unreliable against modern browsers, JA4 is current. See [Proxy & Firewall Log Triage](#).

KQL (Kusto Query Language) — the query language used in Defender XDR's Advanced Hunting. See [Advanced Hunting with KQL](#).

LOLBin (Living-off-the-Land Binary) — a legitimate, signed Windows binary abused for malicious purposes (e.g., `rundll32.exe`, `comsvcs.dll`). See [LSASS Memory Analysis](#).

- MDE (Microsoft Defender for Endpoint)** — the cloud-backed EDR component of the Defender suite, formerly Windows Defender ATP. See [Microsoft Defender Suite](#).
- MDI (Microsoft Defender for Identity)** — monitors on-prem AD signals for identity attacks. See [Defender for Identity Mapping](#).
- MFT (Master File Table)** — the NTFS structure indexing every file and directory on a volume. See [\\$MFT & Timestamping](#).
- NRT (Near Real-Time)** — a Defender custom detection rule frequency running continuously rather than on a fixed schedule. See [Detection Engineering](#).
- NTLM relay** — forwarding a captured NTLM authentication attempt to a different service to authenticate as the victim, without cracking the credential. Referenced in [AD CS Abuse](#) (ESC8).
- PICERL** — SANS's six-phase incident response model: Preparation, Identification, Containment, Eradication, Recovery, Lessons Learned. See [IR Lifecycle](#).
- Plaso / log2timeline** — the standard tool for merging multiple artifact sources into one normalized "super timeline." See [Timeline Construction & Correlation](#).
- PPL (Protected Process Light)** — a Windows protection mechanism that, applied to `lsass.exe`, raises the bar against direct memory-read credential theft. See [LSASS Memory Analysis](#).
- Shimcache / AppCompatCache** — a registry-based cache of executables Windows evaluated for compatibility shimming; proves presence, not confirmed execution, on Windows 10/11. See [Shimcache](#).
- SSP (Security Support Provider)** — a DLL loaded into `lsass.exe` to handle an authentication package; a rogue SSP is a durable credential-theft persistence mechanism. See [LSA Provider / SSP Abuse](#).
- TGT / TGS** — Kerberos Ticket Granting Ticket / Ticket Granting Service ticket. See [Golden Ticket & Silver Ticket Indicators](#).
- TTP (Tactics, Techniques, and Procedures)** — the highest-durability level of threat intelligence, per the [Pyramid of Pain](#).
- UAL (Unified Audit Log)** — Microsoft Purview's cross-workload audit log, with meaningfully longer retention but real ingestion latency compared to Entra ID's native logs. See [Sign-In Logs vs. Audit Logs](#).
- VAD (Virtual Address Descriptor)** — the structure describing a process's memory regions and their permissions; what `vadinfo` / `malfind` actually read. See [Injected Code Detection](#).
- VSS (Volume Shadow Copy Service)** — Windows' point-in-time volume snapshot mechanism, useful for recovering pre-tampering data. See [Volume Shadow Copy Recovery](#).
- XDR (Extended Detection and Response)** — Microsoft's cross-product correlation layer unifying Defender for Endpoint, Identity, Office 365, and Cloud Apps. See [Microsoft Defender Suite](#).
- Zone.Identifier** — the alternate data stream Windows automatically writes to downloaded files (the "Mark of the Web"), recording download origin. See [Alternate Data Streams](#).

2.1 Referenced From

- [Enterprise DFIR Field Guide](#)